

Journal of Transportation Safety & Security

ISSN: 1943-9962 (Print) 1943-9970 (Online) Journal homepage: www.tandfonline.com/journals/utss20

Optimal rail system design with multiple layers of fault and event trees

Chun-Lin Lu & Yung-Cheng (Rex) Lai

To cite this article: Chun-Lin Lu & Yung-Cheng (Rex) Lai (2021) Optimal rail system design with multiple layers of fault and event trees, Journal of Transportation Safety & Security, 13:10, 1135-1156, DOI: 10.1080/19439962.2019.1680584

To link to this article: https://doi.org/10.1080/19439962.2019.1680584

Published online: 30 Oct 2019.

Submit your article to this journal

Article views: 113

View related articles

View Crossmark data

Citing articles: 1 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=utss20

JOURNAL OF TRANSPORTATION SAFETY & SECURITY
2021, VOL. 13, NO. 10, 1135–1156
https://doi.org/10.1080/19439962.2019.1680584

Taylor & Francis
Taylor & Francis Group

Check for updates

Optimal rail system design with multiple layers of fault and event trees

Chun-Lin Lu and Yung-Cheng (Rex) Lai

Department of Civil Engineering, Railway Technology Research Center, National Taiwan University, Taipei, Taiwan

ABSTRACT

A rail system comprises several subsystems and corresponding components. Each element has its life cycle cost (LCC), reliability, and consequences of failure (as described by historical data or an event tree). Determining the optimal rail system design reveals different trends in these characteristics. This research develops an optimization process using two modules to assist the operator in deciding such optimal design. The conversion module first transforms multiple fault tree layers into a simple structure and avoids nonlinear formulation in the optimization model. Then, the optimal system design module identifies the optimal investment plan for rail systems according to available alternatives. Two empirical cases demonstrate the applicability of the proposed optimization process. The first case proves that the conversion module efficiently transforms a fault tree structure. Note that considering data uncertainty into the failure rate requires additional budget allocation to reduce delay costs under ideal and practical situations. The second case indicates that the optimization model can solve a multi-objective problem while considering the LCC and consequences of failure, as well as assist the operator in deciding the appropriate requirement according to their demand.

KEYWORDS

Rail transportation; system design; fault tree; event tree; data uncertainty

1. Introduction

Rail system design is the process of determining the apportionment of system requirements by identifying and procuring the appropriate products for constructing a rail system (Anderson & CoMET, 2006; Anderson, Condry, Findlay, Brage-Ardao, & Li, 2013). This process is usually supported by using existing reliability- and cost-oriented allocation methodologies (Guo, Rong, Yao, & Wang, 2013; Lai, Lu, & Hsu, 2015; Li et al., 2015; Mettas, 2000; Moreb, 2007; Ravi, Reddy, & Zimmermann, 2000). The main concept of these methodologies is to determine an optimal

CONTACT Yung-Cheng (Rex) Lai yclai@ntu.edu.tw Railway Technology Research Center, Department of Civil Engineering, National Taiwan University, Room 313, Civil Engineering Building, No. 1, Roosevelt Road, Sec 4, Taipei 10617, Taiwan.

Color versions of one or more of the figures in the article can be found online at www.tandfonline.com/utss.

© 2019 Taylor & Francis Group, LLC and The University of Tennessee

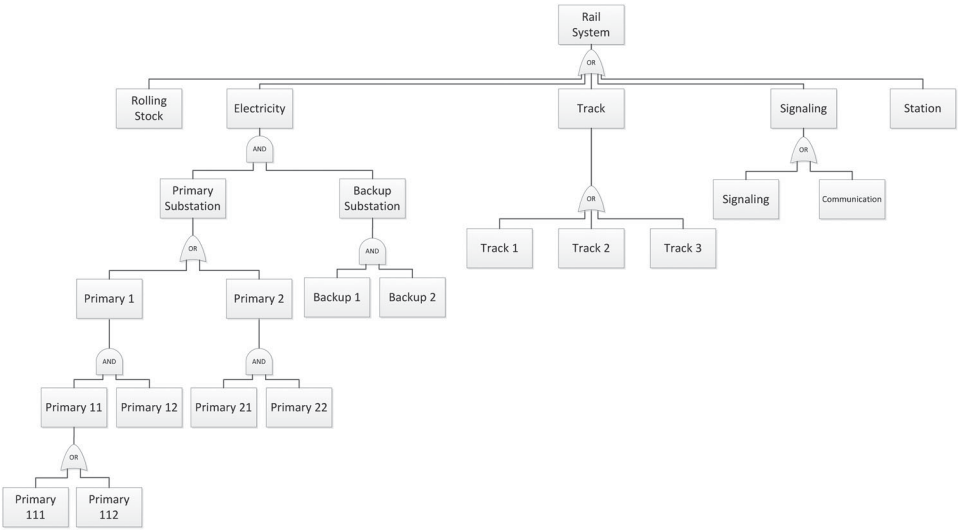


Figure 1. Exemplar fault tree with multiple layers.

investment plan by carefully adjusting the tradeoff between reliability and budget according to available alternatives (Azaiez & Bier, 2007; Elegbede, Chu, Adjallah, & Yalaoui, 2003; Tillman, Hwang, & Kuo, 1997). In this case, an alternative represents a possible product for a specific component.

Most existing studies consider less than three layers of the reliability allocation problem, which usually involves a series–parallel or parallel–series relationship (Burton & Howard, 1969; Gutjahr, Pflug, & Ruszczyński, 1996; Sun, Ruan, & Li, 2006; Yalaoui, Châtelet, & Chu, 2005). These models cannot handle problems with more than three layers (Figure 1) because the possible combinations of alternatives become intractable (due to the number of possible combinations). Another deficiency of prior studies is the inflexibility of the mathematical formulation of their models. The reliability of the rail systems is generally evaluated on the basis of failure rate, mean time between failure (MTBF), or mean distance between failure (MDBF). With this type of attributes, the number of failures is measurable within a specific exposure rate, such as in terms of time (e.g., train-hour) or distance (e.g., train-km). However, this kind of reliability, which can be regarded as "system reliability," fails to consider the effect of failure on trains or passengers (e.g., delay) (Lai, Lu, & Lu, 2017). Lai et al. (2015) proposed the use "service reliability" in the reliability allocation process and developed corresponding optimization models for passenger rail system design. Although service reliability (e.g., delay and on-time percentage) was incorporated in the process, the models are still incapable of handling systems with multiple level structures. In addition, the objectives

of the models from Lai et al. (2015) are only limited to cost and service reliability.

A rail system typically comprises several subsystems and corresponding components. Each element has its life cycle cost (LCC), reliability, and consequences of failure (as described by historical data or an event tree). The event tree (Ballantyne, 1998) can be used to define the characteristics of the consequences of each failure mode in the fault tree. Previous studies on railway system design also failed to examine the uncertainty in the reliability data, but the accuracy of the reliability information acquired from providers could be a key to identifying an optimal investment plan (Lai et al., 2017). For example, field data on a particular component would be more reliable than the reliability data obtained from a laboratory (Coit, Jin, & Wattanapongsakorn, 2004; Rubinstein, Levitin, Lisnianski, & Ben-Haim, 1997). The higher the inaccuracy, the higher the necessity to allocate a buffer for the system design (Allella, Chiodo, & Lauria, 2005; Coit et al., 2004; Marseguerra, Zio, Podofillini, & Coit, 2005). Consequently, this research developed a flexible optimization framework for railway system design allocation by incorporating a conversion process to handle multiple layers of fault and event trees.

2. Railway system design

A fault tree is commonly used to describe the relationship between components and subsystems in a rail system. These relationships would influence the type of failures and produce different failure modes. Numerous products for each component can be selected from several equipment suppliers after introducing a fault tree. Each product has its own characteristics, which separately involve a failure rate, a LCC, and consequences of failure. The failure rate of rail systems is usually estimated on the basis of its failure frequency. The number of failures is measured within a specific exposure rate, such as distance (e.g., train-km), or time (e.g., train-hour). Considerable failures result in high failure rate and suggest that the system is unreliable. LCC can be estimated by using all the cost components for the specific component or system throughout its life. The consequences of failure, which can be evaluated using the historical data from the operator and/or event tree, are also important considerations in the system design. Generally, historical data from the operator directly provide the estimated effect for each consequence, and their effects can be assessed in terms of cost or delay. The effect of the consequences of failure could be incorporated into the rail system design according to these data.

An event tree can also be used to define the characteristic of the consequences of each failure mode in the fault tree. For example, Figure 2

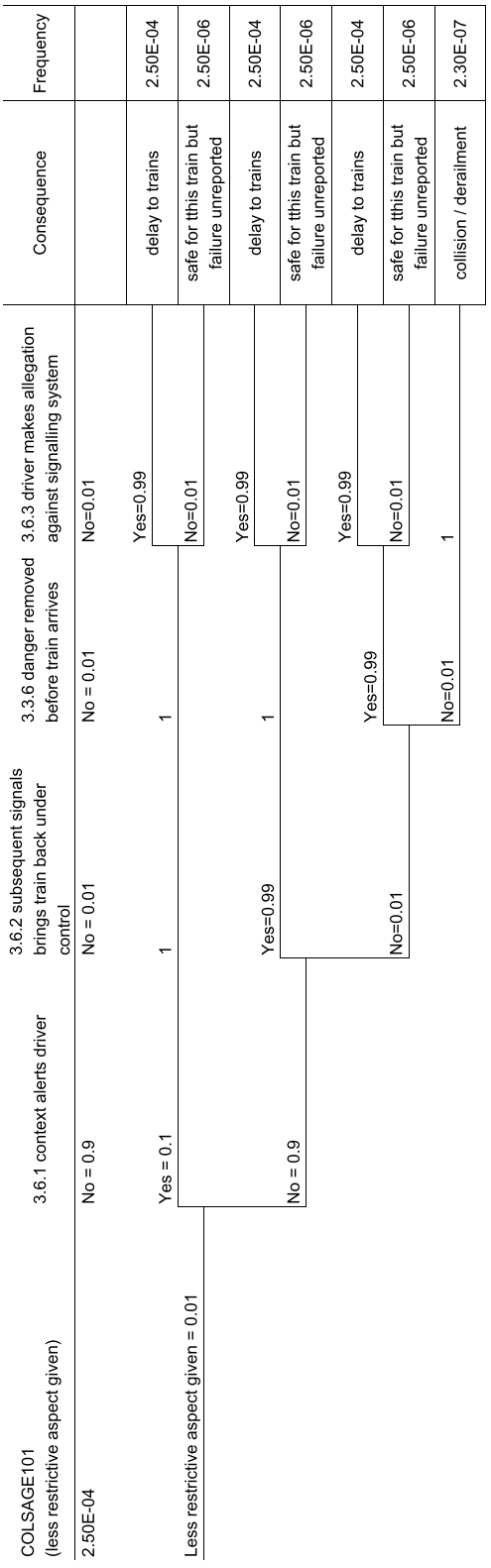


Figure 2. Exemplar event tree (Ballantyne, 1998).

illustrates different barriers (e.g., 3.2.1 "Fault repaired before train stopped at signal") and consequences of failure according to one of the failure modes. Each barrier has its own probability, and the probabilities of each consequence are determined by the relevant barrier. Several kinds of consequences exist; and each one is evaluated using three kinds of factors, namely, delay, repair, and safety (Figure 2). These characteristics are calculated or obtained from the data provided by suppliers. Considering that identifying the optimal rail system design entails different trends produced by various characteristics, the operator must carefully examine each alternative for the components when allocating these characteristics to achieve optimal design. In addition, the operator must establish the related requirement or other demands on the system. These requirements include the selection for purchasing, budget for the LCC, the failure rate of a system, and the effect on the consequences of failure. Various selections are necessary when designing a rail system according to different requirements.

3. Methodology

An optimal railway system design identifies the best alternative for every subsystem or components according to acceptable LCC or consequences of failure. The LCC refers to the summation of costs, which includes construction, renewals, operation, maintenance, and end-of-life costs throughout the product life. Several types of system failure costs, such as expected delay cost and expected safety cost, should also be included in the decision process for system design depending on the preferences of the decision maker. These failures may be attributed to fatigue, man-made damage, or environmental factors. Regardless of the cause of such failures, the decision maker must obtain an understanding about the effects of these occurrences on the rail system. Such reasons would cause rail system failure. Thus, an indicator must be in place to reflect the chances and severity of failure. Accordingly, the expected cost of failure is introduced to determine the effect of failures. The computation of this element requires evaluation via two main factors: failure rate and consequences.

Failure rate can be denoted by μ, which is the reciprocal of the MTBF or MDBF for a product ($\mu = 1/\text{MTBF}$ or $\mu = 1/\text{MDBF}$). A low failure rate improves system reliability. This factor may also become the initial parameter for the calculation of cost for each consequence of failure, and the value of the failure rate may involve various kinds of uncertainty from different data sources. Different consequences have various effects on passengers (waste of time from waiting for the delayed train or suffering casualties in an accident) and operators (repairing the system after an

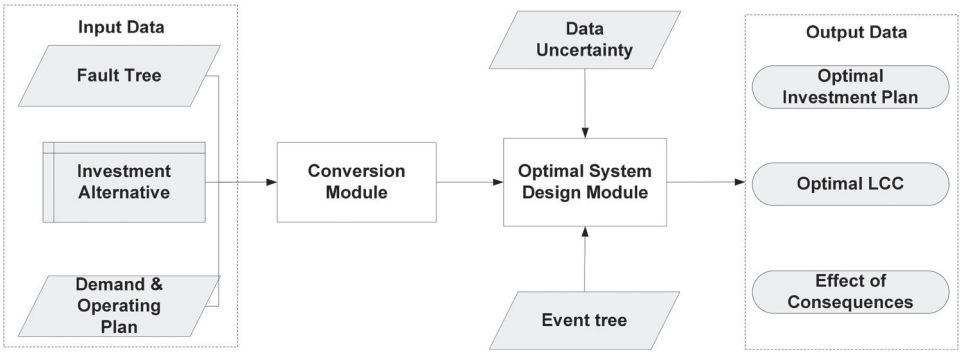


Figure 3. Decision support framework for passenger rail system design.

accident). Therefore, a general and flexible model is necessary to incorporate the possible types of consequences.

Figure 3 presents the optimization process for a rail system design. The conversion module can transform the multiple layers of a fault tree into a simple structure and avoid nonlinear formulation in the optimization model. The optimal system design module aims to determine the optimal investment plan for rail systems based on the available alternatives.

3.1. Conversion module

The increasing complexity of the fault tree layers and their relationship significantly decrease the efficiency in solving the original model. A conversion process is needed to (1) transform multiple fault tree layers into a simple and straightforward structure that is constructed by several failure modes to reduce structure complexity; and, to (2) avoid nonlinear constraints in the optimization model caused by parallel relationships in non-basic component layers, which do not represent the lowest layer of a particular subsystem.

Given these objectives, the process of a conversion module (which is modified from the concept of minimum cut set algorithm) (Pham, 2003) is developed to transform the fault tree structure. Table 1 lists the notations used in the following conversion process, and the subsequent sections discuss the details of the process, as well as demonstrate an exemplar structure with the conversion process.

Phase 1: Setting the Initial Condition

This phase gathers the input data, checks the relationships among different layers, and verifies if an "AND" relationship is present.

Step 1: Start to record relationship (R_j^c) in the joint ($j \in J^l$) from the layer $l \in L^t$ and prepare the set of "OR" (J^s) and "AND" (J^P) relationships.

Table 1. Notation in the minimum cut set algorithm.

Notation	Description
j	Index of joint $j \in J$ (e.g., In Figure 1, "Rail System (Failure)" is the joint of all the five subsystems; Primary 11 is the joint of Primary 111 and Primary 112)
l	Index of layer $l \in L$, (e.g., Figure 1 has six layers from top, namely, the Rail System to Primary 111 and Primary 112 at the bottom)
R_j^c	Relationship (OR/AND) of joint j and its subcomponents
N_j^c	Number of affiliated joints for joint j before conversion
F_j^c	Number of failure modes for joint j after conversion
A_j^c	Set of the affiliated joints of joint j (e.g. Primary Substation and Backup Substation are two affiliated joints of joint "Electricity")
M_j^c	Set of failure modes of joint j
M_t^c	Set of failure modes of the whole system
J	Set of joint j
J^C	Subset of J, representing the converted joints
J^I	Subset of J, representing the joints that constitute a basic component, which is the component in the lowest layer within a subsystem.
J^s	Subset of J, representing the joints with an "OR" relationship with subsequent elements
J^P	Subset of J, representing the joints with an "AND" relationship with subsequent elements
J^l	Subset of J, representing the joints in the layer l
L	Set of layer l (e.g., Layer 1 is the system layer, Layer 2 is the subsystem layer, and so on)
L^t	Subset of L, representing the top layer in the fault tree

Step 2: For each layer $l \in L/L^t$, record the relationship (R_j^c) of the joints ($j \in J^I$) and categorize them into sets of "OR" (J^s) or "AND" (J^P) relationships according to R_j^c.

Step 3: Check the joints $j \in J^P$ to determine the process of conversion.

Step 3.1: If all joints $j \in J^P$ also belong to the set of the joint, namely, the basic components ($j \in J^I$), then this fault tree structure does not require conversion. Proceed to Phase 3.

Step 3.2: If any joint $j \in J^P$ does not belong to the set of the joint in J^I, then assign the relationship (R_j^c) to each joint j and move to Phase 2.

This phase first searches the OR/AND relationship for each layer from the fault tree and records them into the relationship list. The recording would begin from the top layer, which is represented by the rail system in Figure 1. In the relationship list (Table 2), the name of the joint (rail system) and the relationship between this layer and the affiliated joint ("OR" relationship) would be recorded. After recording the data of the top layer, the affiliated joint would be analyzed until all joints and layers have been recorded in the relationship list. If a non-basic component layer exists in the "AND" relationship (e.g., joint "Electricity"), then the combination of alternatives would increase sharply. These joints would be transformed in Phase 2 to simplify the system structure.

Phase 2: Converting the Fault Tree Structure and Data Form

This phase changes the fault tree structure and converts the form of the "AND" relationship into a non-basic component layer. The fault tree structure would be converted into several failure modes.

Table 2. Relationship list for the structure in Figure 1.

Joint	Relationship	Layer
Rail system	OR	Layer 1
Electricity	AND	Layer 2
Track	OR	Layer 2
Signaling	OR	Layer 2
Primary substation	OR	Layer 3
Backup substation	AND	Layer 3
Primary 1	AND	Layer 4
Primary 2	AND	Layer 4
Primary 11	OR	Layer 5

Step 1: For all joints $j \in J$, record the affiliated joints into A_j^c, which is a set of the affiliated joints of joint j.

Step 2: For all joints $j \in J$, calculate the number of joints in A_j^c and assign it as N_j^c, which is the number of affiliated joints for joint j before conversion.

Step 3: For all joints $j \in J$, check their relationship R_j^c for the type of conversion and assign joint j into J^C.

> ***Step 3.1:*** If the relationship of joint R_j^c is "OR," then assign the N_j^c as the number of failure modes for joint j after conversion (F_j^c). Then, arrange every joint $j \in A_j^c$ into M_j^c and proceed to Step 4.
>
> ***Step 3.2:*** If the relationship of joint R_j^c is "AND," then assign 1 as the number of failure modes for joint j after conversion (F_j^c) and arrange this joint j into M_j^c. Then, proceed to Step 4.

Step 4: For all joints $j \in J$, check that the conversion process has been executed, which means that N_j^c and $M^c{}_j$ have been assigned and $J^C = J$.

> ***Step 4.1:*** If all joints $j \in J$ have been arranged to failure mode ($J^C = J$), then proceed to Phase 3.
>
> ***Step 4.2:*** If any existing joint $j \in J$ has not been arranged to a failure mode (that is, any $j \notin J^C$), then repeat Step 3.

Step 1 generates Table 3 to demonstrate the relationship between joint and affiliated joint. The number of affiliated joints is also computed in Step 2. According to the relationship list from Table 3, the conversion process begins with the top layer (i.e., the rail system) and checks the relationship of the joints layer by layer in Steps 3 and 4. If the joint is in an "OR" relationship, then all affiliated joints represent a specific failure mode; as a result, this joint would produce the same number of failure modes as the number of affiliated joints. For example, if a rail system has five affiliated joints, then the conversion for this layer transforms them into five failure modes. If the joint is in an "AND" relationship, then all affiliated joints represent only one failure mode; as a result, this joint would produce only one failure mode. In another example, an electricity subsystem with two affiliated joints (i.e., the primary and backup substations) indicates that the conversion for this layer transforms them into one failure mode.

Table 3. Relationship list with affiliation condition for the structure in Figure 1.

Joint	Relationship	Affiliated to	# of affiliated joints
Rail system	OR	–	5
Electricity	AND	Rail system	2
Track	OR	Rail system	3
Signaling	OR	Rail system	2
Primary substation	OR	Electricity	2
Backup substation	AND	Electricity	2
Primary1	AND	Primary substation	2
Primary2	AND	Primary substation	2
Primary11	OR	Primary1	2

Phase 3: Producing the Final Data List for the Optimization Model

After converting the data form and fault tree structure, this phase generates the list of failure modes and produces the final data form structure.

Step 1: For all $j \in J$, assign the failure mode in set M_j^c into the set of the failure mode of the entire system M_t^c.

Step 2: For $j \in J$ in Step 3.1 of Phase 1, assign j into M_t^c.

Step 3: For $j \in J^l$ and $l \in L^t$, assign the new "OR" relationship to the top layer in R_j^c.

After the conversion from Phase 2, the 10 resulting failure modes include: (1) Rolling Stock, (2) Primary Substation 111 – Primary Substation 12 – Backup Substation 1 – Backup Substation 2, (3) Primary Substation 112 – Primary Substation 12 – Backup Substation 1 – Backup Substation 2, (4) Primary Substation 21 – Primary Substation 22 – Backup Substation 1 – Backup Substation 2, (5) Track 1, (6) Track 2, (7) Track 3, (8) Signaling, (9) Communication, and (10) Station. Phase 3 produces the converted fault tree and lists all the failure modes (Figure 4).

3.2. Optimal system design module

The optimization module aims to decide the optimal investment plan for rail systems according to available alternatives. As mentioned, three main considerations exists in passenger rail system design: multiple layers with different relationships, the data uncertainty on failure rate, and the consequences of failures. The optimal investment plan identifies the best alternative for every component according to the decision maker's requirement through the minimization of the LCC and all types of expected cost of failures. The basic formulation of this module is similar to that in Lai et al. (2017). However, we extend and improve the model to enable it to incorporate all possible consequences of failures (as specified by the users), allocate the appropriate buffer considering data uncertainty, and handle any fault and/or event trees. Given that multiple types of consequences can

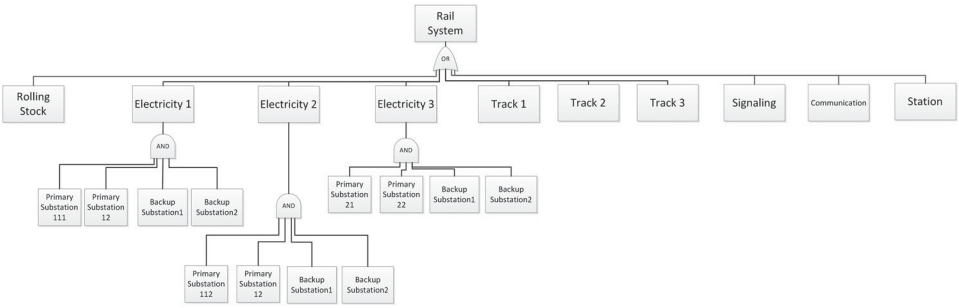


Figure 4. Exemplar fault tree with multiple layers after conversion.

simultaneously exist with the occurrence of failure, the objective function must incorporate the cost of different types of consequences.

The following notations are used in the optimization module. i is the index of the component (Figure 1). j is the index of the joint (subsystem, sub-subsystem …). n is the index of the alternatives of the component. λ is index of the consequences of failure. I, J, N, and Λ are the sets of i, j, n, and λ, respectively. k is the index of the combinations of alternatives within a subsystem j. K_j is the set of k within joint j. J^r is the subset of J, that represents all the subsystems. J^B is the subset of J that denotes all subsystems. J^s is a subset of J that represents the subsystem with an "OR" relationship with subsequent elements. J^P is the subset of J that denotes the subsystem with an "AND" relationship with subsequent elements. V_k is the set of corresponding alternatives in combination with k, where $(i, n) \in V_K$. N_i is the set of alternatives of i elements. L is the set of joint pairs with a connecting relationship (e.g., the track subsystem and Track 1 component in Figure 1). C_{in} is the LCC of alternative n of component i. Q_j^{λ} is cost of consequence λ of joint j (e.g., the subsystem or failure mode). D_j is the average delay per failure for subsystem j. $F^C{}_{in}$ is the failure rate of alternative n of component i. U_j is the number of elements in j subsystem. P is the total operational time per year. R is design service reliability. B is the design LCC. E is the design system reliability for the system. G_j is the design system reliability in terms of the failure rate of each subsystem j. S^{λ} is the system requirement for consequence λ. γ_{in} is the multiplied factor for the failure rate of the alternative n of component i in the highest buffer (risk averse) consideration. H is the planning horizon of the project. m is a large negative number (e.g., -1×10^8).

The five types of decision variables are used in the optimization model:

- δ_{in} is a binary variable that denotes the selection of the alternative, where $\delta_{in} = \begin{cases} 1, & \textit{if alternative } n \textit{ of component } i \textit{ is selected} \\ 0, & \textit{otherwise} \end{cases}$.

- y_{jk} is a binary variable that denotes the selection of the combination, where

$$y_{jk} = \begin{cases} 1, & \textit{if combination } k \textit{ of joint } j \textit{ is selected} \\ 0, & \textit{otherwise} \end{cases}.$$

- f_{jk}^{C} is a continuous variable that denotes the failure rate of each combination k in joint j.
- f_{j}^{B} is a continuous variable that denotes the failure rate of joint j.
- f is a continuous variable that denotes the failure rate of the system.

The following formulations are applied in the optimization models.

Objective Function

$$Min\left(\sum_{i\in I}\sum_{n\in N} C_{in}\delta_{in} + H\sum_{\lambda\in\Lambda}\sum_{j\in J_B} Q_j^{\lambda} f_j^{B}\right), \tag{1}$$

Objective function (1) minimizes the total LCC and expected failure costs. The LCC refers to the summation of the costs of all components in the design process, including construction, renewals, operation, maintenance, and end-of-life costs. Total failure costs refer to the summation of the costs of each consequence multiplied by the design life span. Each consequence of failure has its own cost with the occurrence of failure, and this value may vary in different modes. We introduce a parameter Q_j^{λ}, the cost of consequence λ of joint j, to capture the effect for each consequence by incorporating the preceding equations into rail system design. Thus, the effect of every consequence can be ascertained after the supplier or decision-maker conducted examination or evaluation.

Alternative Selection Constraints

$$\sum_{n\in N_i}\delta_{in} = 1 \qquad \forall i \in I, N_i \in N, \tag{2}$$

$$\sum_{k\in K} y_{jk} = 1 \qquad \forall j \in J_r, \tag{3}$$

$$\sum_{(i,n)\in V_k}\delta_{in} \le y_{jk} + (U_j - 1) \quad \forall k \in K_j, j \in J_r, \tag{4}$$

Equation (2) guarantees a one-to-one relationship between alternatives and components. Each component can only appear once in a rail system because the fault tree restricts the type of components. As a result, only one of all alternatives of one component is selected. Equation (3) guarantees that only one combination (k) can be selected for a joint. Equation (4) maintains the relationship between the combination and corresponding alternatives for all components in the joint, indicating that a combination

only appears when all the alternatives of the component in this combination are selected.

Failure Rate Computation Constraints

$$f_{jk}^{C} \geq F_{in}^{C}\gamma_{in} + m(1-y_{jk}) \qquad \forall (i,n) \in V_k, k \in K_j, j \in J_s, \tag{5}$$

$$f_{jk}^{C} \geq \prod_{(i,n)\in V_k} F_{in}^{C}\gamma_{in} + m(1-y_{jk}) \qquad \forall k \in K_j, j \in J_p, \tag{6}$$

Equations (5) and (6) compute the failure rates of the combination k of joint j that consists of components with an "OR" or "AND" relationship, respectively. Equation (5) determines the failure rate of combination k of joint j (i.e., f^{C}_{jk}) that consists of components with an "OR" relationship, which shows the highest failure rate among all components i and the corresponding alternatives n within the subsystem. $1\text{-}y_{jk}$ indicates that each alternative of the components would return a failure rate only when the combination contains this alternative of the component. Equation (6) determines the failure rate of combination k of joint j (i.e., f^{C}_{jk}) that consists of components with an "AND" relationship, which returns the failure rate of joint j as the product of the failure rate of all components i within the subsystem.

Data uncertainty occurs when noisy data cause deviation from the correct, intended, or original values. In rail system design, data uncertainty usually originates from the value of the failure rate (i.e., the reliability of the alternative). For a particular alternative, reliability can be estimated from historical data (if available) or from the suppliers. As a result, according to the sources of data and the different abilities on capturing value, failure rate should be evaluated at different levels to indicate the type of data uncertainty. In Equations (5) and (6), γ_{in}, the multiplied factor for the failure rate of alternative n of component i, considers this possible uncertainty.

System Failure Rate Constraints

$$f_j^{B} \geq f_{jk}^{C} \qquad \forall k \in K_j, j \in J_r, \tag{7}$$

$$f_j^{B} \geq f_{j'}^{B} \qquad \forall (j,j') \in L, \tag{8}$$

$$f \geq f_j^{B} \qquad \forall j \in J_r, \tag{9}$$

$$f \leq E, \tag{10}$$

Equation (7) calculates the value of the failure rate of joint j (i.e.,f_j^{B}) from f_{jk}^{C}, thereby allowing the failure rate of the joint combination to become the failure rate of the joint. Equation (8) computes the value of the failure rate of joint j (i.e.,f_j^{B}) from other joints j' (i.e.,$f_{j'}^{B}$), thereby ascertaining the failure rate between layers. Similarly, Equation (9) identifies the failure rate of the system (i.e., f) from f_j^{B} with an "OR" relationship. Equation

(10) guarantees the minimum requirement of the failure rate of the system, as specified by the operator or decision maker.

System Requirement Constraints

$$\sum_{k\in K} f_{jk}^{C} y_{jk} \le G_j \qquad \forall j \in J_B, \tag{11}$$

$$\sum_{i\in I}\sum_{n\in N} C_{in}\delta_{in} \le B, \tag{12}$$

$$\left(\frac{P-\sum_{j\in J_r} D_j f_j^B}{P}\right) \times 100\% \ge R, \tag{13}$$

Equation (11) ensures that the requirement of design system reliability has been covered. Equation (12) ensures that the total LCC of the optimal investment plan is within the LCC design. Equation (13) guarantees that service reliability is greater than design service reliability. Users can determine these requirements according to each specific system.

Event Tree Constraint

$$\sum_{j\in J_B} Q_j^{\lambda} f_j^B \le S^{\lambda} \qquad \forall \lambda \in \Lambda, \tag{14}$$

Equation (14) is developed for a specific requirement of each consequence of failure in a year. Every consequence has its own cost (Q_j^{λ}) when failure occurs, and the requirement S^{λ} can limit them with a design value. Different failure modes indicate diverse consequences of failure. For example, one failure mode may have several kinds of consequences, and each consequence has its own probability of occurrence. Thus, the total consequences for each failure mode should be incorporated in terms of cost after establishing the relationship between FTA and ETA. ETA can generate the cost of consequences, which has been illustrated with the entire life cycle in the objective function. To further consider these consequences of failure, the related requirement can be adopted to reflect the effects from different consequences.

Variable Domain

$$\delta_{in} \in \{0,1\} \qquad \forall i \in I, N_i \in N, \tag{15}$$

$$y_{jk} \in \{0,1\} \qquad \forall j \in J_r, k \in K, \tag{16}$$

$$f_{jk}^{C} \ge 0 \qquad \forall k \in K_j, j \in J_r, \tag{17}$$

$$f_j^B \ge 0 \qquad \forall j \in J, \tag{18}$$

$$f \ge 0. \tag{19}$$

Equations (15) to (19) describe the types of decision variables. The decisions on the selection of alternatives of components and combinations are binary variables. For this optimization model, the variables for calculating

the failure rate of combinations of alternatives, joints, and systems are positive variables.

4. Case study

Two case studies were performed using all the developed modules to demonstrate their applicability. Case I identifies the advantage of using a conversion module and provides a comparison with the effect of data uncertainty introduction. Case II uses detailed fault and event trees for the signal subsystem to demonstrate the consideration of multiple consequences of failure.

4.1. Case I: consideration of data uncertainty

A 25 km passenger rail system in Taiwan was considered in Case I to demonstrate the application of the proposed optimization process. Figure 1 shows the structure of the system, subsystems, and subsequent components. Each component has numerous alternatives, such as suppliers and products, with specific reliability, LCC, and impact on consequences of failure.

The ridership forecast was 140,000 trips/day. The planning horizon was set at 30 years. Figure 5 depicts the information on the possible alternatives for all subsystems and components. The LCC generally increases with the decrease in failure rate, but the relationship is nonlinear.

In this case, the impact of data uncertainty and comparison with conversion modules were introduced. The multiplied factor was proposed to reflect different levels and identify various kinds of data uncertainty. This factor is categorized into three levels based on the sources of data. The first level is based on historical data, which can be obtained from a similar system; the two other levels involve estimated data from the supplier, with a few tests before operation. The second level can be estimated with low data uncertainty because they have been tested by independent verification and validation (IV&V), as performed by a third-party organization that is not involved in product development. The third level should entail high data uncertainty because these data were exclusively tested by the suppliers themselves. According to different levels and scenarios, the multiplied factor for Level 1 uses 1.15 as the value because this level involves historical data with high credibility. Given that Level 2 has IV&V to prove its credibility and Level 3 has none, the values of the multiplied factors for Levels 2 and 3 are estimated as 1.4 and 2, respectively.

Figure 1 illustrates the original fault tree structure of the project. If each component has 10 alternatives, then the number of combinations of the electricity subsystem would become 10 million. This situation would cause

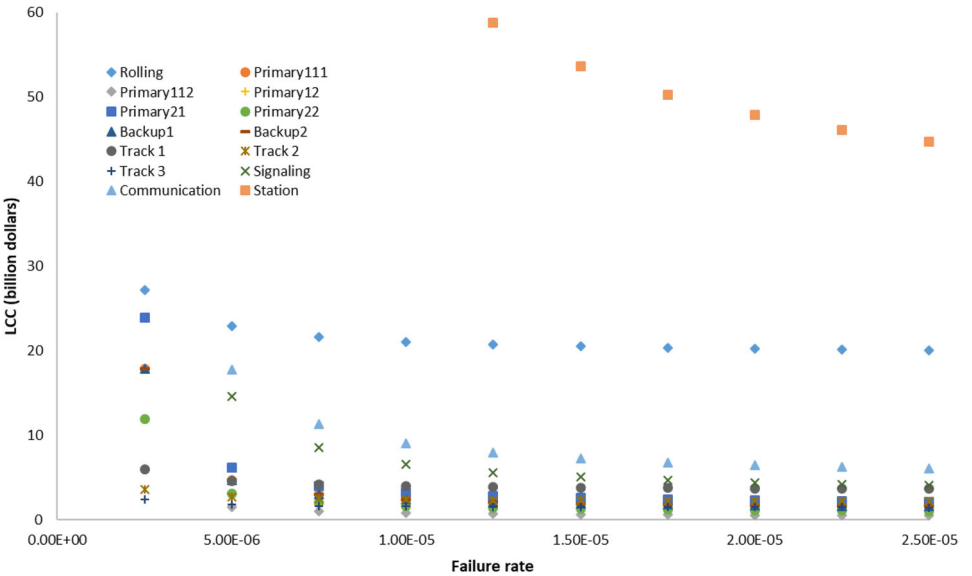


Figure 5. Alternatives for new system design.

extreme difficulty for our general model to identify the best alternatives for each component. Consequently, the minimum cut set algorithm was used to reduce the complexity of the fault tree.

Moreover, the consequences of failure in this case only consider the delay time (D_j), which is the product of the operational time (T_j), the average number of online trains (N), and the average delay time (Q_j). A result of 5,411,347 (train-km/year) was obtained, with an average of 31 online trains. Q_j was estimated according to the historical data from existing systems that exhibit characteristics similar to those of the new design. The average time value per person was approximately 2.95 NTD/person-minute, and the average number of passengers per train was 96 (Department of Rapid Transit Systems, 2005). The resulting time value (P^T) was 16,992 NTD/train-hour. Delay cost is the product of expected delay time (the product of failure rate and D_j) and design time value (P^T). After computing D_j, the LCC, system reliability (in failure rate), and consequences of failure of all alternatives could be inputted into the optimization models to determine the optimal investment plan.

Three sub-cases were proposed to compare these conditions using the same evaluator. Case I-I represents the condition without the conversion process. Case I-II describes the condition after the conversion process. Given that Cases I-I and I-II assume that all data are certain, Case I-III is designed to consider data uncertainty. All optimization models were established on the basis of the "Optimal System Design Module" for each situation as solved using CPLEX, which is a popular commercial solver for linear programing problems developed by ILOG (2009). Table 4 lists the

Table 4. Solution efficiency between Cases I-I and I-II.

Type	# of var.	# of const.	Solution time (sec)	Objective value (billion NTD)	Improvement rate of solution time
Case I-I	22,357,349	41,578,628	172,800	95.927	97.7%
Case I-II	93,519	159,839	4,060	95.927	

resulting efficiency and quality between Cases I-I and I-II for the comparison of their specific differences. The numbers of variables and constraints decrease to approximately thousandths with the conversion module, and the solution time decreases by approximately 97.7% compared with its no-conversion counterpart. Thus, the conversion module effectively transforms the fault tree structure. Table 4 also indicates that both situations can identify the optimal solution, thereby suggesting that the conversion module can solve the problem while maintaining the quality of the solution. Accordingly, the two sub-cases prove that the conversion module has the advantage of handling multiple layers from the fault tree.

Two scenarios were designed for different data uncertainty conditions to illustrate the benefit of considering data uncertainty. The first one is the ideal scenario, in which the alternatives function under the condition of ideal reliability). The other one is the "practical scenario," where the alternatives function under the condition in which reliability would be 1.15, 1.4, or 2 times away from the ideal condition according to different uncertainty levels. Table 5 shows the results for each subsystem in different sub-cases. The same budget constraint, 94 billion NTD, was used for all models. The total LCCs of the optimal investment plans for all conditions (which have less than 0.6% difference among them) are similar. However, the values of the delay costs show obvious differences among the three sub-cases, and the ideal and practical situations present dissimilar results for these cases. The total delay cost would decrease by approximately 10.79% with data uncertainty when the situation is ideal. However, if the situation is practical, then the total delay cost would decrease by approximately 31.81% with data uncertainty, a value which is more than that in the ideal situation. This finding is explained by the following facts: Cases I-I and I-II regard failure rate as an ideal value without any uncertainty, thereby ignoring the effect of data uncertainty on the failure rate. By contrast, Case I-III considers data uncertainty into the failure rate and must allocate an additional budget to reduce the delay cost under ideal and practical situations.

4.2. Case II: investment selection with consideration of the event tree

In this case, more than one consequence of failure, as described by the event tree, would be considered in the rail system design. The fault and event trees for the signaling subsystem in Proctor (1997) and Ballantyne

Table 5. Optimal investment plan in ideal and practical scenarios.

			Delay cost (billion NTD)			
	LCC (billion NTD)		Ideal		Practical	
Subsystem	Case I-I and I-II	Case I-III	Case I-I and I-II	Case I-III	Case I-I and I-II	Case I-III
Rolling stock	20.058	20.058	0.147	0.147	0.169	0.169
Electricity	10.863	10.912	0.100	0.090	0.315	0.233
Track	7.491	7.792	0.510	0.241	1.15	0.367
Signaling	10.181	10.363	1.588	1.588	2.557	2.001
Stations	44.749	44.749	0.240	0.240	0.276	0.276
Total	93.342	93.874	2.585	2.306	4.467	3.046
Percent difference	0.57%		−10.79%		−31.81%	

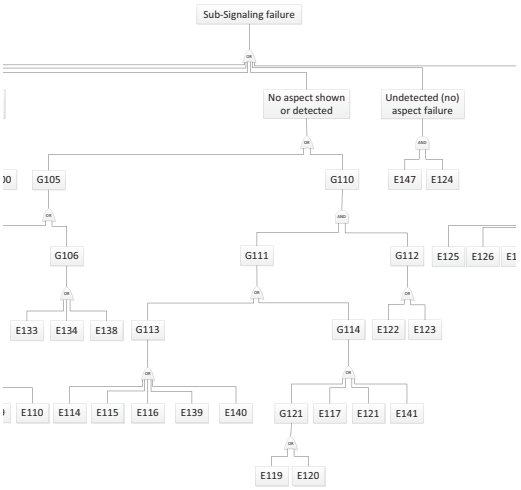
(1998) were used for this case study. Figure 6a illustrates the fault tree of the signaling subsystem from the report, which contains six failure modes ("less restrictive aspect given," "no aspect shown or detected," "undetected [no] aspect failure," "signal cannot be clearly seen," "aspect fails to danger," and "failure of the release mechanism") and 42 components. Each component has 10 alternatives with specific failure rates and LCCs.

The event tree (Figure 6b) in the report provides the related barrier and the types of consequences for each failure mode. Each scenario was evaluated by three main consequences: repair, delay, and safety. The calculation of the delay cost has been proposed in the previous section, and the repair cost was provided by the report (Ballantyne, 1998). The total safety cost for one accident can be calculated through Equation (20), where C_A is the value of life in a railway accident, for which 7.9 million NTD is used as recommended by the Institute of Transportation (Suen, 2013). N_{mj} is the injured situation, and $E^F{}_m$ is the equivalent fatality (EQF). The corresponding EQF values for major and minor injuries are 0.1 and 0.01, respectively.

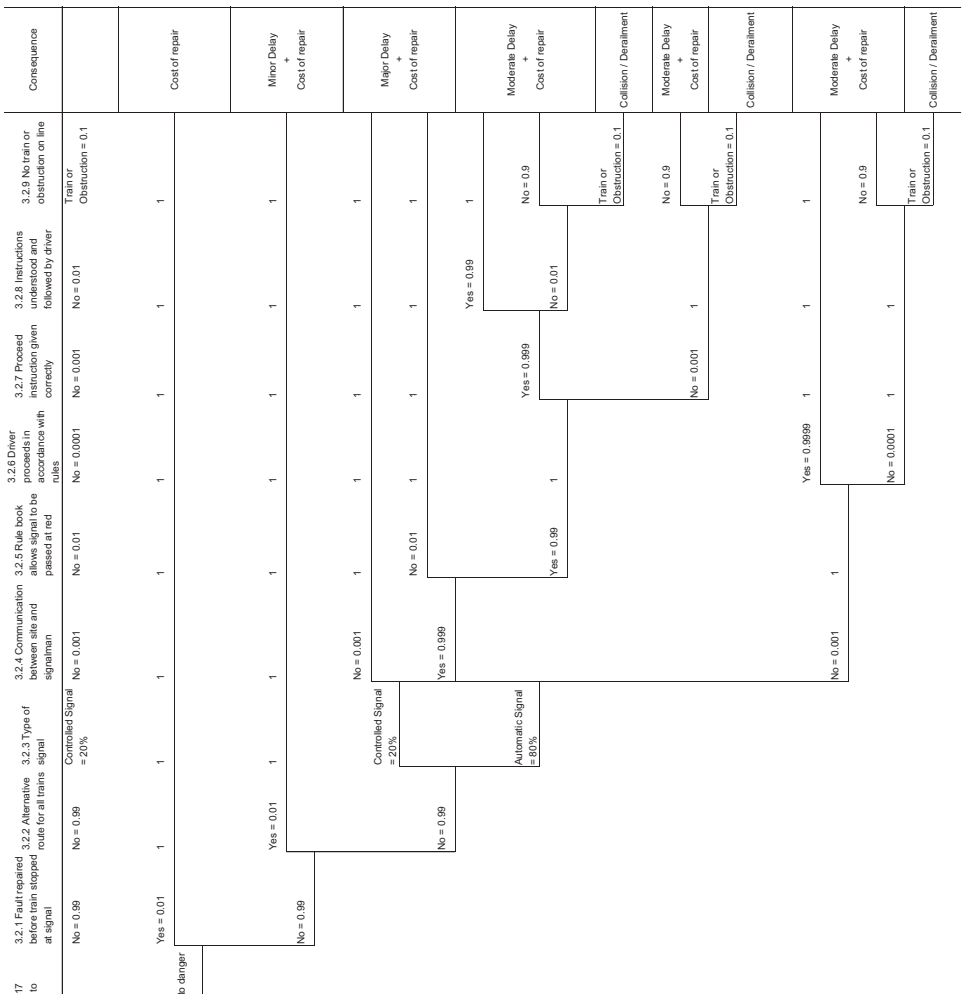
$$Q_j^A = C_A \sum_m N_{mj} E_m^F, \qquad \forall j \in J. \tag{20}$$

This case is again solved within seconds by using CPLEX. Table 6 shows the optimal allocation with consideration of the event tree determined by the optimization model. The resulting LCC was \$4.612 billion NTD, the total delay cost of the planning horizon was \$1.701 billion NTD, the total repair cost was \$1.150 million NTD, and total safety cost was \$0.269 million NTD. This allocation demonstrates the optimal balance between LCC and several consequences of failure at a given time value. The LCC and delay cost occupy most values of the objective value, which means that the other consequences have minimal influence on the rail system.

In addition, the requirement for each consequence could influence the investment plan. In this case, we selected safety as the representative consequence because delay has been discussed in prior studies (e.g., Lai et al., 2015, 2017). Moreover, the requirement for repair is not intuitive for the operator. By contrast, Table 6 indicates that the consequence from safety is



(a)



(b)

Figure 6. (a) Fault tree and (b) event tree of the signaling subsystem (Proctor, 1997 & Ballantyne, 1998).

Table 6. Optimal investment plan.

	LCC	Delay	Repair	Safety
Cost (billion NTD)	4.612	1.701	1.15E-03	2.69E-04
Percent of objective value	73.041%	26.939%	0.016%	0.004%

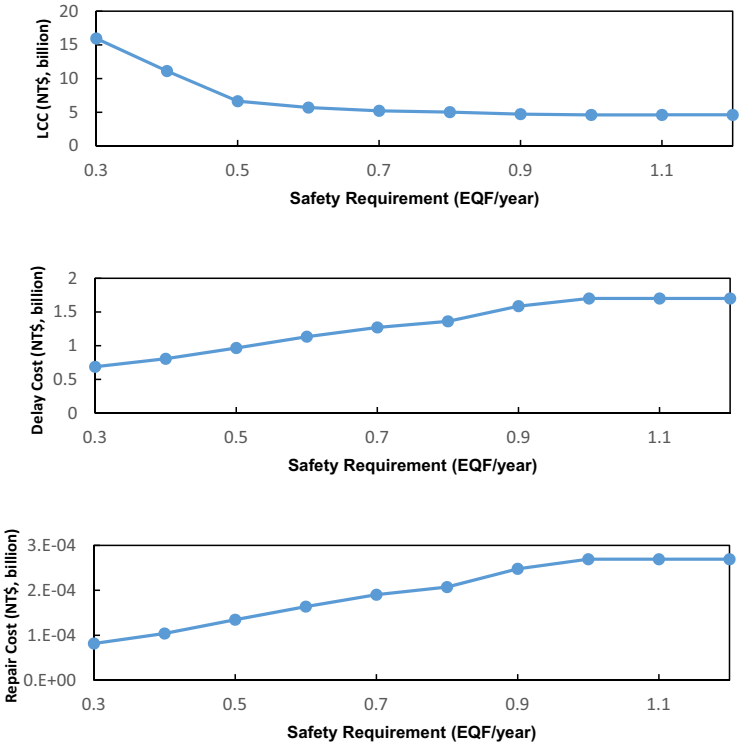


Figure 7. Scenario analysis for safety requirement.

approximately 0.972 EQF/year, which is equal to 1.87E − 9 EQF/person-km/year. This value is better than the requirement for the MRT in Singapore (2.20E − 9 EQF/person-km/year) and Hong Kong (7.50E − 9 EQF/person-km/year), thereby suggesting that the optimization model can consider the minimum requirement for safety. A scenario analysis was performed by examining the impact of different requirements on safety cost that varied from 0.3 EQF/year to 1.2 EQF/year to assist the operator in identifying an appropriate requirement for safety.

In this scenario analysis, 1 EQF/year (9,217 NTD/year) represents the softest requirement annually. Total safety cost without any requirement for this case (Table 6) was 269,000 NTD. Thus, the safety loss in a year was 8,967 NTD at most. Consequently, the requirement with 1.1–1.2 EQF will not change the selection of the operator. Conversely, 0.3 EQF/year represents the strictest requirement for each year. For scenarios with less than 0.5 EQF/year, LCC sharply increases (Figure 7) with the decrease in

allowable EQF per year. The delay and repair costs all decrease as the requirement for safety becomes strict because these costs are all attributed to failure. According to this analysis, if the operator looks for an improved safety level, then setting the requirement to anywhere above 0.5 EQF/year seems to be affordable, given that the LCC remains in the similar range. A requirement anywhere below 0.5 EQF/year would mean a substantial increase in LCC.

5. Conclusion

This study develops an optimization process using two modules, namely, conversion and optimal system, to assist the operator in the appropriate selection of rail system design. The conversion module can transform multiple layers of a fault tree into a simple structure and avoid nonlinear formulation in the optimization model, while the optimal system design module aims to determine the optimal investment plan for rail systems according to available alternatives. This optimization process can identify the best alternative for every subsystem according to acceptable LCC or consequences of failure. Two empirical cases were investigated using the developed models to demonstrate their applicability. The first case proves that the conversion module can effectively transform the fault tree structure. Moreover, considering data uncertainty in the failure rate requires additional budget allocation to reduce delay costs under ideal and practical situations. The second case confirms that the optimization model can solve the multi-objective problem while considering LCC and the consequences of failure, as well as assist the operator in determining the appropriate requirement according to their demand.

Funding

This project was funded by the Ministry of Science and Technology (MOST) of Taiwan under Grant No. 104-2221-E-002 -204 -MY5.

References

Allella, F., Chiodo, E., & Lauria, D. (2005). Optimal reliability allocation under uncertain conditions, with application to hybrid electric vehicle design. *International Journal of Quality & Reliability Management*, *22*(6), 626–641. doi:10.1108/02656710510604926

Anderson, R., Condry, B., Findlay, N., Brage-Ardao, R., & Li, H. (2013). *Measuring and valuing convenience and service quality: A review of global practices and challenges from mass transit operators and railway industries* (International Transport Forum Discussion Paper No. 2013/16). Paris: Organisation for Economic Co-Operation and Development.

Anderson, R. N. & CoMET. (2006). *Improving the management and delivery of mass public transportation in cities*. Presented at Nova Symposium, Buenos Aires.

Azaiez, M. N., & Bier, V. M. (2007). Optimal resource allocation for security in reliability systems. *European Journal of Operational Research*, *181*(2), 773–786. doi:10.1016/j.ejor.2006.03.057

Ballantyne, T. R. (1998). Infrastructure Risk Modelling Electrical Signals - Multiple Aspect Main Type (Consequence Model) (Railtrack EE&CS Report, No. RT/S&S/IRM_CCA/15). London, UK: Railtrack.

Burton, R. M., & Howard, G. T. (1969). Optimal system reliability for a mixed series and parallel structure. *Journal of Mathematical Analysis and Applications*, *28*(2), 370–382. doi:10.1016/0022-247X(69)90036-5

Coit, D. W., Jin, T., & Wattanapongsakorn, N. (2004). System optimization with component reliability estimation uncertainty: A multi-criteria approach. *IEEE Transactions on Reliability*, *53*(3), 369–380. doi:10.1109/TR.2004.833312

Department of Rapid Transit Systems. (2005). *Feasibility study of north-south MRT route for eastern Taipei area*. Presented at Taipei City Government, Taipei City.

Elegbede, A. O. C., Chu, C., Adjallah, K. H., & Yalaoui, F. (2003). Reliability allocation through cost minimization. *IEEE Transactions on Reliability*, *52*(1), 106–111.

Guo, S., Rong, Z., Yao, J., & Wang, H. (2013). Reliability modeling and assigning method for HXD electric locomotive. In Proceedings of the International Conference on Quality, Reliability, Risk, Maintenance, and Safety Engineering (QR2MSE), Chengdu, China, pp. 289–295.

Gutjahr, W. J., Pflug, G. C., & Ruszczyński, A. (1996). Configurations of series-parallel networks with maximum reliability. *Microelectronics Reliability*, *36*(2), 247–253. doi:10.1016/0026-2714(95)00004-L

Ilog, I. (2009). *IBM ILOG CPLEX V12.1: User's manual for CPLEX*. New York, NY: IBM Corp.

Lai, Y. C., Lu, C. T., & Hsu, Y. W. (2015). Optimal allocation of life-cycle cost, system reliability, and service reliability in passenger rail system design. *Transportation Research Record: Journal of the Transportation Research Board*, *2475*(1), 46–53. doi:10.3141/2475-06

Lai, Y. C., Lu, C. T., & Lu, C. L. (2017). A comprehensive approach to allocate reliability and cost in passenger rail system design. *Transportation Research Record: Journal of the Transportation Research Board*, *2608*(1), 86–95. doi:10.3141/2608-10

Li, W., Qin, Y., Lin, S., Jia, L., An, M., Zhang, Z., … Li, H. (2015). Reliability allocation of high-speed train bogie system. In Proceedings of the International Conference on Electrical and Information Technologies for Rail Transportation (pp. 609–617). Berlin Heidelberg: Springer.

Marseguerra, M., Zio, E., Podofillini, L., & Coit, D. W. (2005). Optimal design of reliable network systems in presence of uncertainty. *IEEE Transactions on Reliability*, *54*(2), 243–253. doi:10.1109/TR.2005.847279

Mettas, A. (2000). Reliability allocation and optimization for complex systems. In *Proceedings of the Annual Reliability and Maintainability Symposium* (pp. 216–221). Los Angeles, CA, USA. https://ieeexplore.ieee.org/abstract/document/816310

Moreb, A. A. (2007). Allocating repairable system's reliability subject to minimal total cost – An integer programming approach. *Journal of Systems Science and Systems Engineering*, *16*(4), 499–506. doi:10.1007/s11518-007-5060-8

Pham, H. (2003). *Handbook of reliability engineering*. London: Springer.

Proctor, P. (1997). Infrastructure Risk Modelling Electrical Signals - Multiple Aspect Main Type. (Railtrack EE&CS Report, No. RT/S&S/IRM_FTA/15). London, UK: Railtrack.

Ravi, V., Reddy, P. J., & Zimmermann, H. J. (2000). Fuzzy global optimization of complex system reliability. *IEEE Transactions on Fuzzy Systems*, *8*(3), 241–248. doi:10.1109/91.855914

Rubinstein, R. Y., Levitin, G., Lisnianski, A., & Ben-Haim, H. (1997). redundancy optimization of static series-parallel reliability models under uncertainty. *IEEE Transactions on Reliability*, *46*(4), 503–511. doi:10.1109/24.693783

Suen, C. S., Lin, D. H., Li, C. K., Chang, K. K., & Wu, X. Z. (2013). The rail system case of risk assessment in different countries. *Journal of Sinotech*, *118*, 75–85.

Sun, X., Ruan, N., & Li, D. (2006). An efficient algorithm for nonlinear integer programming problems arising in series-parallel reliability systems. *Optimization Methods and Software*, *21*(4), 617–633. doi:10.1080/10556780600627800

Tillman, F. A., Hwang, C. L., & Kuo, W. (1997). Optimization techniques for system reliability with redundancy: A review. *IEEE Transactions on Reliability*, *26*(3), 148–155. doi:10.1109/TR.1977.5220100

Yalaoui, A., Châtelet, E., & Chu, C. (2005). A new dynamic programming method for reliability & redundancy allocation in a parallel-series system. *IEEE Transactions on Reliability*, *54*(2), 254–261. doi:10.1109/TR.2005.847270